
Do PLMs Understand Binding Affinity?

Or do we need to learn it?

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Machine Learning for Natural Sciences





Motivation

What PLMs Do Well

PLMs (ESM-2, ProtT5) learn rich representations from millions of sequences. These capture:

Structure: secondary structure, contact maps, 3D coordinates

Function: GO terms, enzyme activity, protein families

Binding sites: [InterPLM discovered SAE features activating on binding-site residues \(Nature Methods 2025\)](#)

CLAPE-SMB showed frozen ESM-2 embeddings can partially distinguish binding from non-binding residues via t-SNE.

The Open Question

Binding site identification is a structural/geometric property of the protein alone.

Binding affinity is fundamentally different:

It is a thermodynamic property of a specific protein-ligand pair (pKd/pKi). Knowing where something can bind is not the same as knowing how strongly a specific ligand binds there.

Do frozen PLM embeddings encode quantitative binding affinity — a pair-level, continuous signal? Or are other methods needed to extract it?



Research Questions

Q1 Do frozen PLM embeddings contain linearly decodable binding affinity signal?

Test via linear probes and shallow MLPs on frozen embeddings for continuous pKd/pKi prediction . No contrastive training, no fine-tuning.

Q2 Does affinity signal localize to the same residues/layers as binding site features?

InterPLM found site features in mid layers. Does affinity require the same or additional information beyond site recognition?

Apply Use probing insights to build a lightweight, informed binding prediction pipeline.

Use optimal layer, residue subset, and minimal architecture from probing. Benchmark vs. ConPlex, BAPULM, CAPLA on PDBbind/KIBA/CASF-2016.



Prior Interpretability Work on PLMs

SAE Probing (Garcia et al., PNAS 2025)

SAEs + transcoders on ESM-2.

Linear probes on SAE embeddings match ESM baselines for structure, localization, and thermostability.

Scope: protein-level properties. Binding affinity not tested.

CLAPE-SMB (Wang et al., J Cheminf 2024)

Frozen ESM-2 + 5-layer MLP for binding site prediction. Uses contrastive learning loss. t-SNE shows partial separability of binding/non-binding residues in raw embeddings.

MCC = 0.529 (SJC), 0.699 (UniProtSMB).

Scope: binary residue-level site classification. No affinity quantification.

InterPLM (Simon & Zou, Nat Methods 2025)

Trained SAEs on ESM-2 embeddings; identified 2,548 interpretable features per layer.

Binding sites among discovered features — PLMs encode site info in superposition.

Scope: binary site identification (structural property of protein alone). Does NOT address affinity for ligands.



Related Work: Extraction Methods

Method	Approach	Key Finding	What It Assumes
ConPLex (Singh et al., PNAS 2023)	Contrastive co-embedding of ProtBert + Morgan fingerprints	Contrastive training increases drug/decoy effect size from 0.73 → 4.72	PLM provides useful features (untested independently)
DrugCLIP (Gao et al., NeurIPS 2023 / Science 2026)	CLIP-style retrieval using 3D pocket encoder (UniMol)	Ultrafast screening: 10T+ pairs in <24h, 15-17.5% wet-lab hit rates	Pocket encoder learns binding from contrastive loss
BindCLIP (arXiv, Feb 2026)	CLIP-style + pocket-conditioned diffusion for pose generation	DrugCLIP representations rely on shortcut correlations, not true binding	Contrastive + generative > contrastive alone
BAPULM (arXiv Nov 2024)	Frozen ProtT5 + MolFormer → feedforward network	R = 0.925 (Benchmark1k), 0.914 (Test2016)	FFN may learn binding itself; PLM signal not isolated
CAPLA (Bioinf 2023)	Cross-attention between pocket and ligand	Attention scores identify critical binding residues	Only after task-specific training

All methods treat PLM/pretrained embeddings as a starting point and optimize downstream. None isolate what the raw PLM already knows.

? The Gap We Address

What Has Been Probed

Binding sites (InterPLM, CLAPE-SMB)

Binary: is this residue part of a binding pocket? A structural property of the protein alone.

Protein-level properties (Garcia et al.)

SAE probes for structure, localization, thermostability — not binding.

What Nobody Has Probed

Binding affinity (our focus)

Continuous, pair-level: given protein X and ligand Y, what is pKd? A thermodynamic property requiring pair representation.

Affinity localization

Does affinity signal concentrate in the same residues/layers that encode binding sites, or does it require additional information?

Our question: *"How much affinity signal exists in frozen PLM embeddings — and can we use it?"*



Proposed Methodology

Stage 1: Probing Analysis

1

Extract Representations

Run sequences through frozen ESM-2 (multiple sizes). Get per-residue + pooled embeddings.

2

Linear Probing (Q1)

Train linear classifiers/regressors on frozen embeddings for binary binding and continuous affinity.

3

Localization (Q2)

Probe subsets: binding-site residues, non-binding residues, random controls. Compare performance.

4

Layer Analysis

Probe across PLM layers — where does binding signal emerge? How does it compare with InterPLM's analysis?

Stage 2: Informed Method

5

Design Lightweight Pipeline

Use probing findings: optimal layer, residue subset, minimal architecture.

6

Benchmark

Compare vs. ConPLex, BAPULM, CAPLA on PDBbind, KIBA, CASF-2016.

7

Controls

Shuffle control, family-stratified splits, ligand-agnostic vs. conditioned.

8

Ablation Study

Show why design choices work — grounded in probing results from Stage 1.



Datasets & Baseline Models

Datasets

Dataset	Purpose	Type
PDBbind 2020 Refined	Primary affinity dataset with 3D structures + binding-site annotations	Continuous (pKd/pKi)
KIBA	Large-scale kinase inhibitor bioactivities	Continuous (KIBA score)
BindingDB	Broad positive binding samples	Continuous
DUD-E	Hard negative (decoy) samples for specificity testing	Binary
CASF-2016	Held-out evaluation benchmark	Continuous

Baseline Models for Comparison

Model	Type	SOTA Metric
ConPLex (Singh et al., 2023)	Contrastive co-embedding (ProtBert)	DUD-E: effect size 4.72
BAPULM (arXiv, 2024)	Frozen PLM + feedforward (ProtT5 + MolFormer)	Test2016: R = 0.914
CAPLA (Jin et al., 2023)	Cross-attention (sequence-level)	Test2016: R = 0.843
SSM-DTA	Current KIBA SOTA	KIBA: CI = 0.894



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